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3D Vision

# Lecture 10: 3D Object Detection, 3D Pose Estimation

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# Contents

# Object Detection

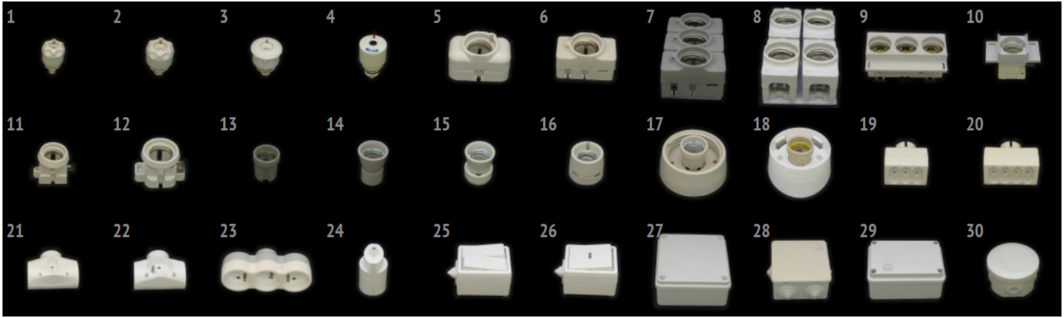
The 3D object detection task has more variants than the standard 2D task on RGB images. In general the goal of the task is to detect 3D positions of objects. Generally there are several different types of this task:

- 6 DoF pose estimation - detect the positions ($R$ and $\boldsymbol{t}$) of a known object
- Categorical object detection - detect a 3D bounding box around an object of a given category
- Bird's eye view detection - detect a 2D bounding boxes of objects in a bird's eye view of the scene

We can also consider multiple different types of input:

- RGB images - one (monocular) or more
- RGB-D or depth maps
- Pointclouds

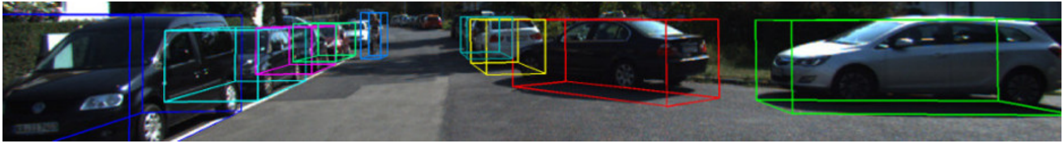
# 6 DoF Pose Estimation




For this task we may have a CAD model available. In some cases the method assumes that the object will be recorded from multiple views prior to detection.

Image adopted from: Tomáš Hodan et al. “T-LESS: An RGB-D dataset for 6D pose estimation of texture-less objects.” In: *2017 IEEE Winter Conference on Applications of Computer Vision (WACV)*. IEEE. 2017, pp. 880–888

# Categorical 3D Object Detection



Each bounding box is defined by its width, height and depth, translation and rotation. In some cases the rotation may be limited fewer than 3 DoF (as in the dataset shown above). There can be multiple categories. There is some ambiguity w.r.t. the correct rotation of some objects.

Image adopted from: Andreas Geiger, Philip Lenz, and Raquel Urtasun. “Are we ready for autonomous driving? the kitti vision benchmark suite.” In: *2012 IEEE conference on computer vision and pattern recognition*. IEEE. 2012, pp. 3354–3361

# Annotating 3D Objects

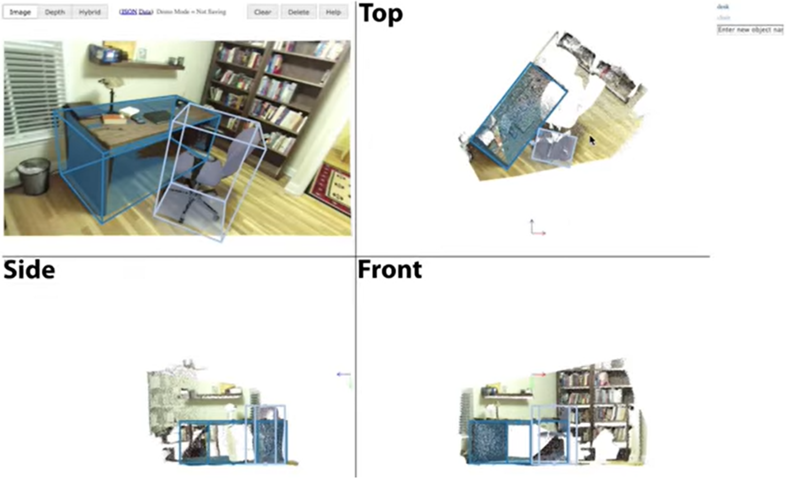



Image adopted from: Shuran Song, Samuel P Lichtenberg, and Jianxiong Xiao. "Sun rgb-d: A rgb-d scene understanding benchmark suite." In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2015, pp. 567–576

# Standard SfM vs Deep Learning

If we have a sequence of images with a given object we can obtain its sparse reconstruction. We can then use this to detect the object. We simply do this by finding the $R$ and $\boldsymbol{t}$ of the object instead of the whole scene using a standard PnP algorithm.

Note that there may still be an issue with the $R$ and $\boldsymbol{t}$ based on the sparse model, which may be aligned in an arbitrary manner (e.g. coordinates of the first view used for reconstruction)

# Regressing Parameters

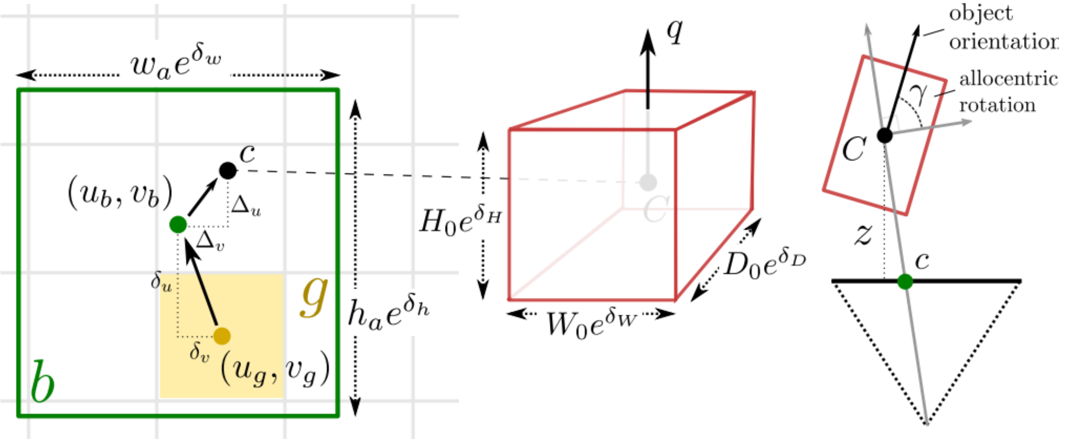
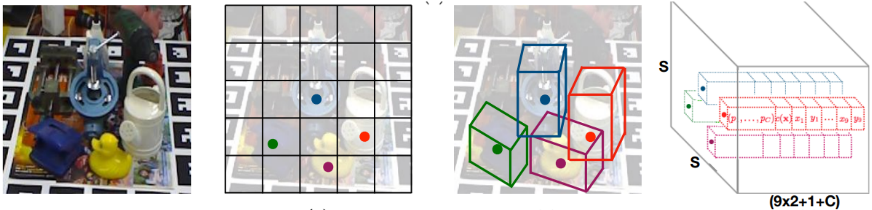



Image adopted from: Andrea Simonelli et al. "Disentangling monocular 3d object detection." In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2019, pp. 1991–1999

# Corners + PnP




(b) (c) (d) (e)

Figure 1. Overview: (a) The proposed CNN architecture. (b) An example input image with four objects. (c) The $S \times S$ grid showing cells responsible for detecting the four objects. (d) Each cell predicts 2D locations of the corners of the projected 3D bounding boxes in the image. (e) The 3D output tensor from our network, which represents for each cell a vector consisting of the 2D corner locations, the class probabilities and a confidence value associated with the prediction.

Image adopted from: Bugra Tekin, Sudipta N Sinha, and Pascal Fua. "Real-time seamless single shot 6d object pose prediction." In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 292–301

# 3D Location of Points + PnP

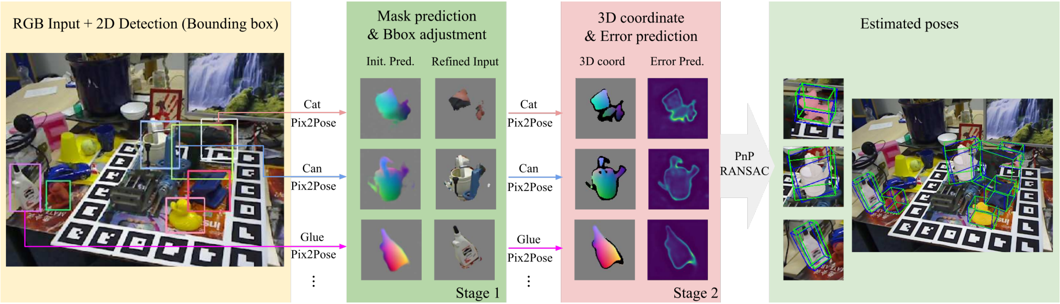



Image adopted from: Kiru Park, Timothy Patten, and Markus Vincze. "Pix2pose: Pixel-wise coordinate regression of objects for 6d pose estimation." In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2019, pp. 7668–7677

# Dealing with Symmetries

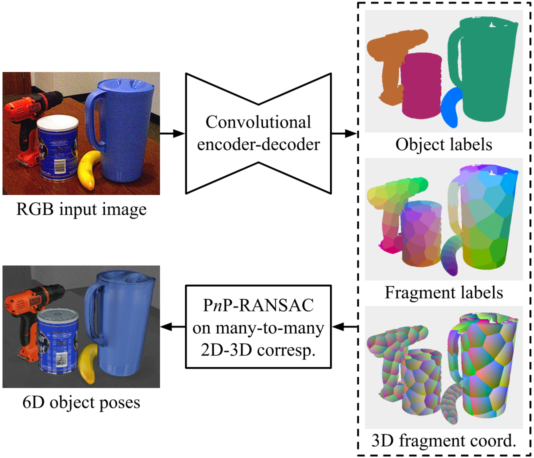



Image adopted from: Tomas Hodan, Daniel Barath, and Jiri Matas. "Epos: Estimating 6d pose of objects with symmetries." In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2020, pp. 11703–11712

# Faster 2D-3D Correspondences

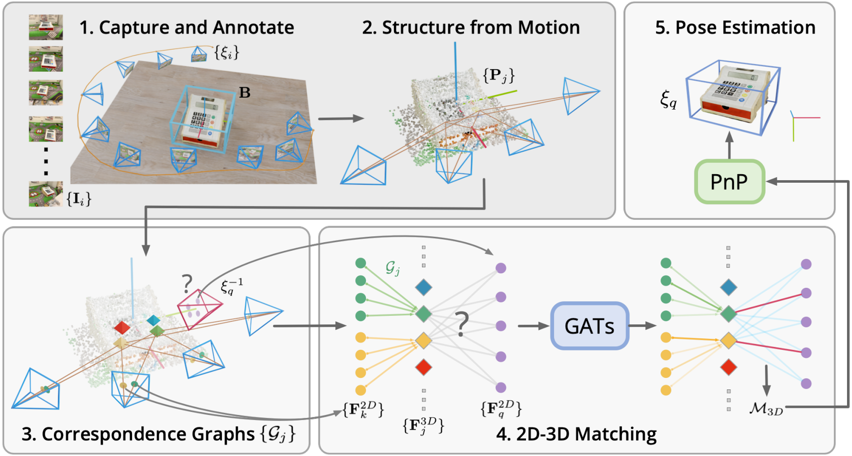



Image adopted from: Jiaming Sun et al. “Onepose: One-shot object pose estimation without cad models.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 6825–6834

# Regressing $R$ and $\boldsymbol{t}$

We may also be interested in regressing $R$ and $\boldsymbol{t}$ for the objects. Regressing $\boldsymbol{t}$ is usually straightforward, but working with $R$ has several problems.

The fundamental issue with regressing $R$ from a neural network is that it has 3 DoF, but there is not *continuous* 3-dimensional representation for $\mathrm{SO}(3)$.

# Representing $R$

If we output simply $R$ out of the network we would have problem in that $R$ won't be in $\mathrm{SO}(3)$. We could project a matrix $\hat{R}$ from a network onto $\mathrm{SO}(3)$ (Procrustes problem), but that is not very common.

Some other approaches include:

- Three (Euler) angles - problems with periodicity, can be solved by binning - regression turns into classification
- Quaternions - both $q$ and $-q$ represent the same rotation, discontinuity near small rotations
- Rodrigues' vector - problem with periodicity, $-\boldsymbol{v}$ vs. $\boldsymbol{v}$, discontinuity near small rotations

# Two Vector Approach

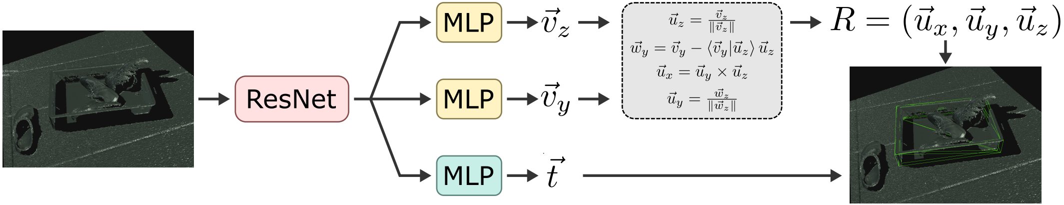



Image adopted from: Lukáš Gajdošech et al. “Towards Deep Learning-based 6D Bin Pose Estimation in 3D Scans.” In: *arXiv preprint arXiv:2112.09598* (2021)

# Symmetries and $R$

Symmetric objects pose a challenge. Let us consider rotations $R_s \in S$ such that rotating the object by $R_s$ keeps the object pose equivalent. In other words $R \sim R_s R$. Note that $I \in S$. Such symmetries can be dealt with by using a loss of the form:

$$\hat{L} = \min_{R_s \in S} L(R_s R, R_{gt}). \tag{1}$$

This loss may have multiple minima or pleateau areas. This can be problematic in terms of training dynamics. It is possible to deal with this directly via paramterization of $R$ or splitting the output to multiple regressors and a classifier.

# Frustum PointNet

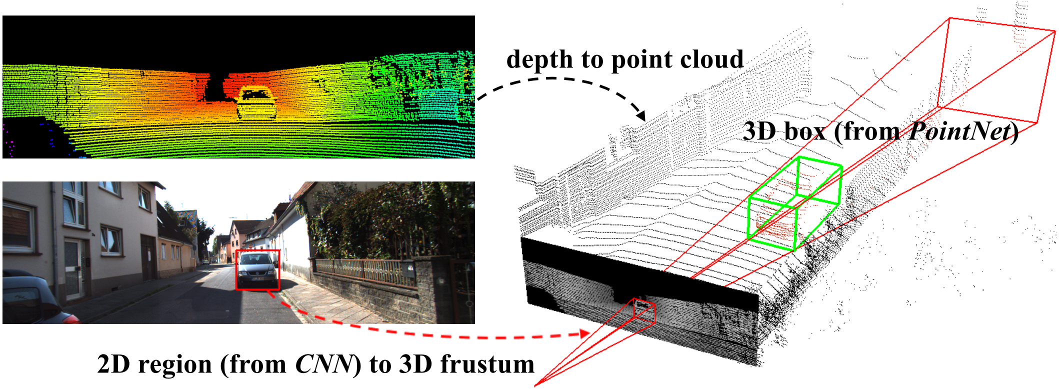



Image adopted from: Charles R Qi et al. “Frustum pointnets for 3d object detection from rgb-d data.” In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 918–927

# Frustum PointNet

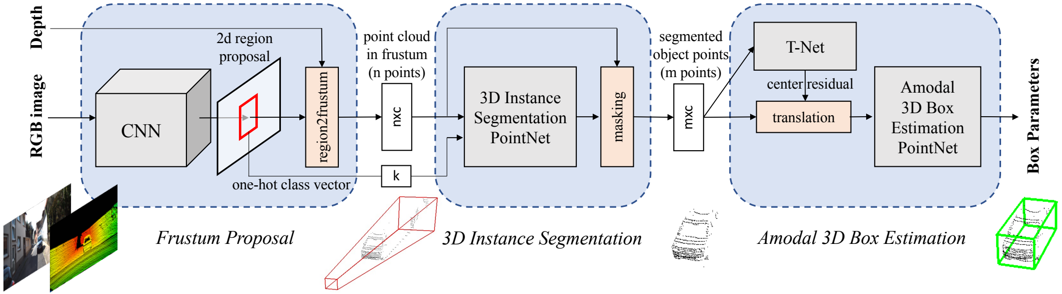



Image adopted from: Charles R Qi et al. "Frustum pointnets for 3d object detection from rgb-d data." In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018, pp. 918–927

# Point Pillars

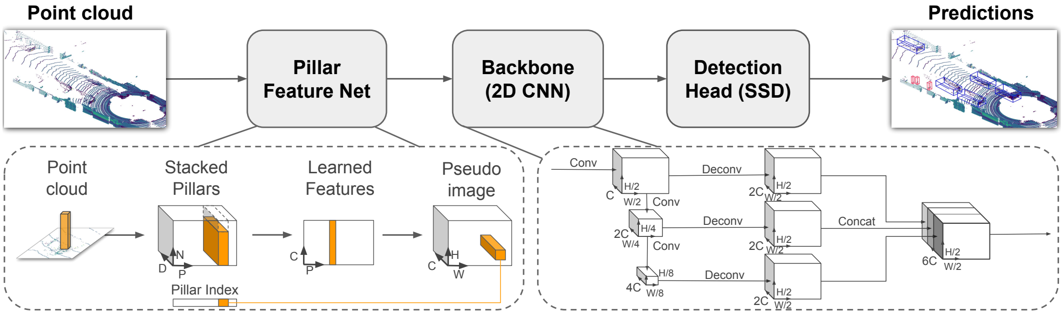



Image adopted from: Alex H Lang et al. “Pointpillars: Fast encoders for object detection from point clouds.” In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2019, pp. 12697–12705

# Pseudolidar

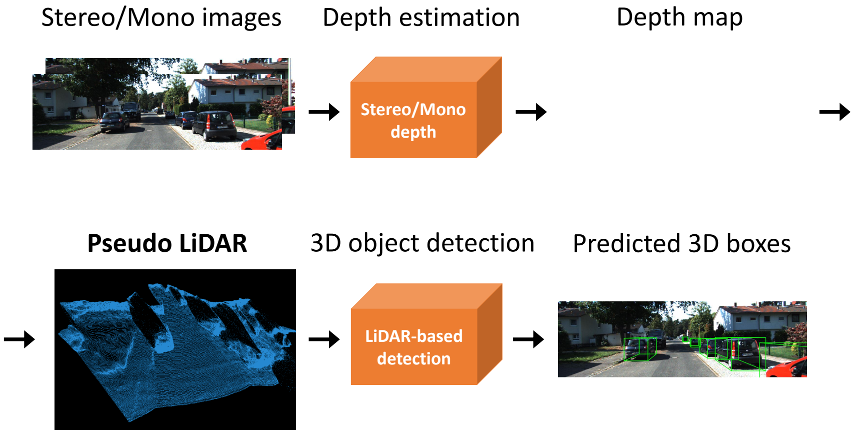



Image adopted from: Yan Wang et al. “Pseudo-lidar from visual depth estimation: Bridging the gap in 3d object detection for autonomous driving.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 8445–8453

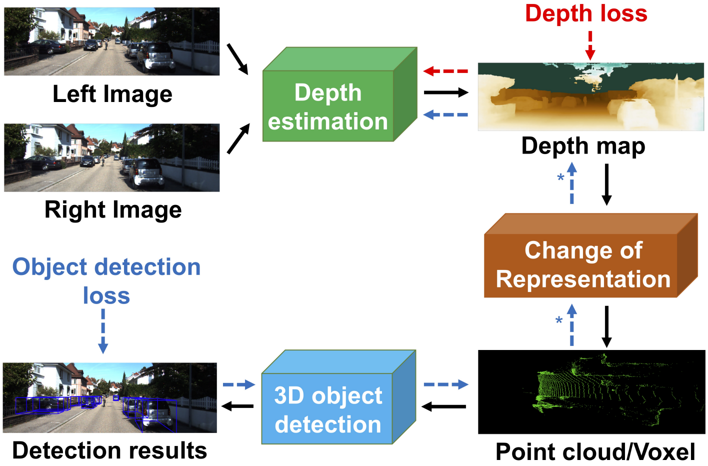



Image adopted from: Rui Qian et al. “End-to-end pseudo-lidar for image-based 3d object detection.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2020, pp. 5881–5890

# Depth-aware Transformers

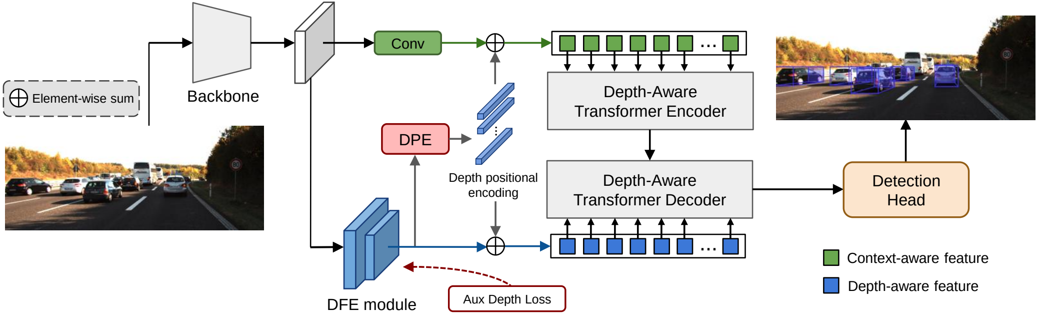



Image adopted from: Kuan-Chih Huang et al. “Monodtr: Monocular 3d object detection with depth-aware transformer.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022, pp. 4012–4021

# Sensor Fusion

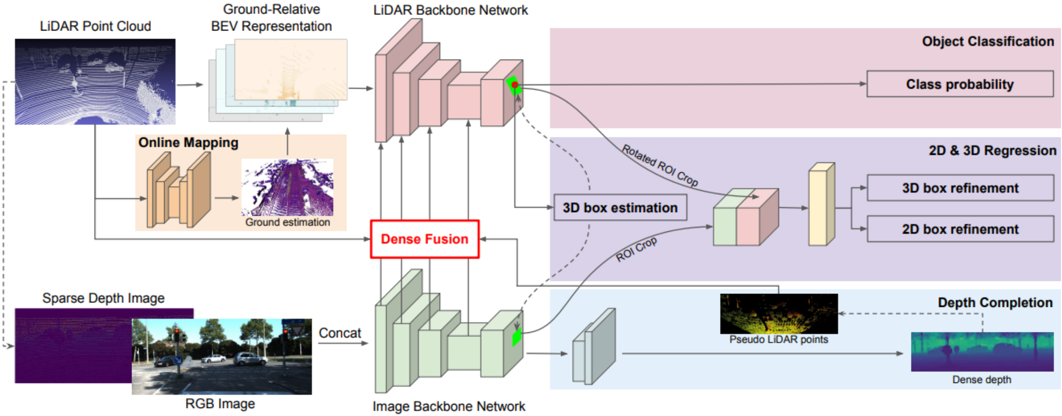



Image adopted from: Ming Liang et al. “Multi-task multi-sensor fusion for 3d object detection.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 7345–7353

# BundleSDF

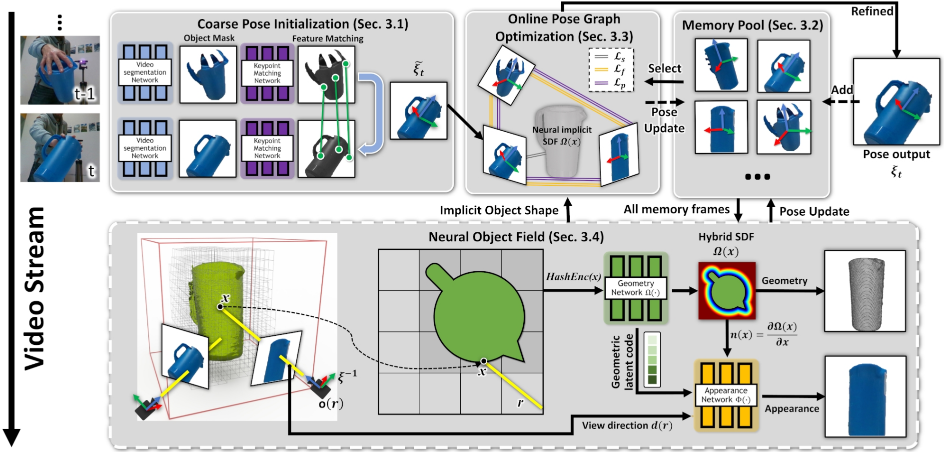



Image adopted from: Bowen Wen et al. “Bundlesdf: Neural 6-dof tracking and 3d reconstruction of unknown objects.” In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023, pp. 606–617

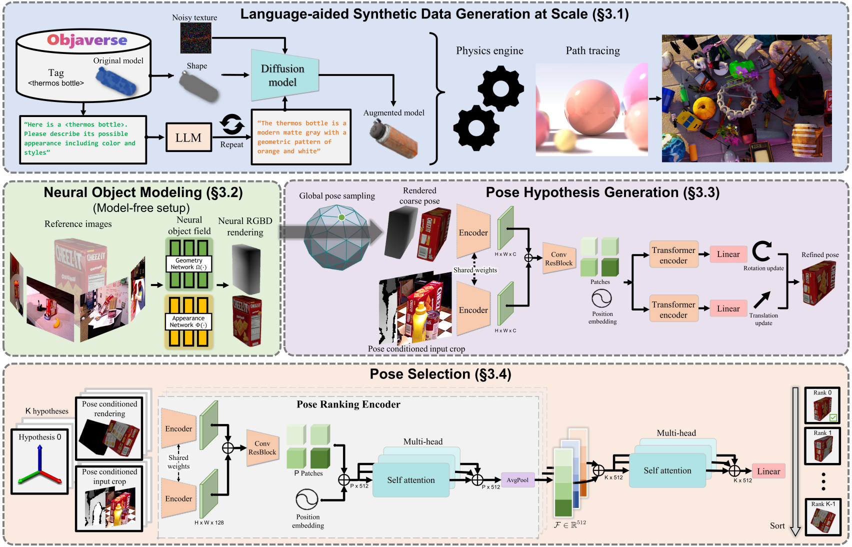



Image adopted from: Bowen Wen et al. "Foundationpose: Unified 6d pose estimation and tracking of novel objects." In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024, pp. 17868–17879

# Human Pose Estimation 2D to 3D

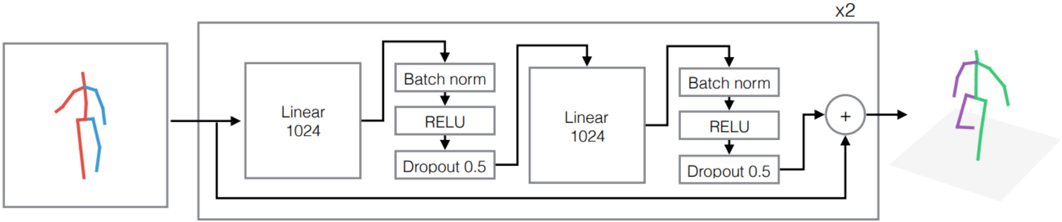



Image adopted from: Julieta Martinez et al. "A simple yet effective baseline for 3d human pose estimation." In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 2640–2649

# Weakly-supervised Human Pose Estimation

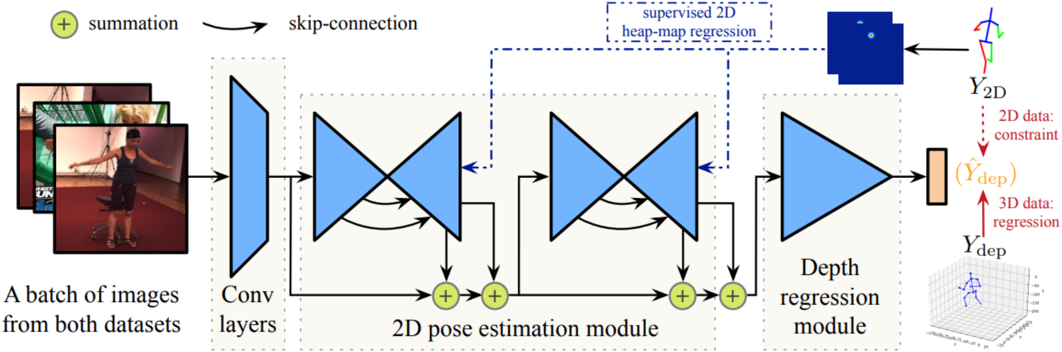



Image adopted from: Xingyi Zhou et al. "Towards 3d human pose estimation in the wild: a weakly-supervised approach." In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 398–407

# 3D Hand Pose Estimation

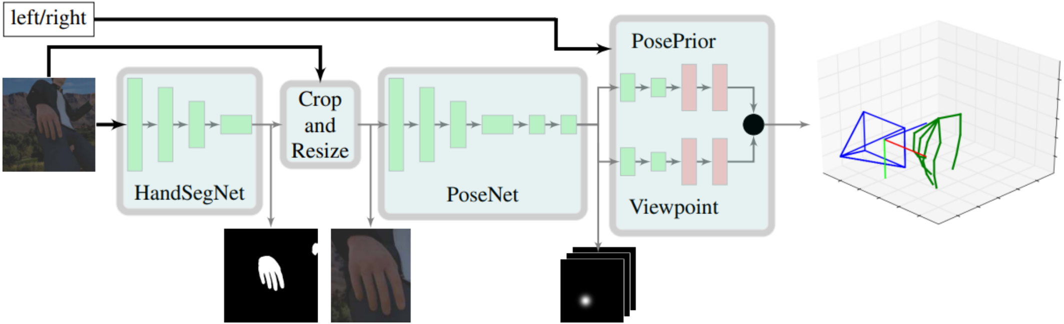



Image adopted from: Christian Zimmermann and Thomas Brox. "Learning to estimate 3d hand pose from single rgb images." In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 4903–4911

# 3D Hand PosePrior

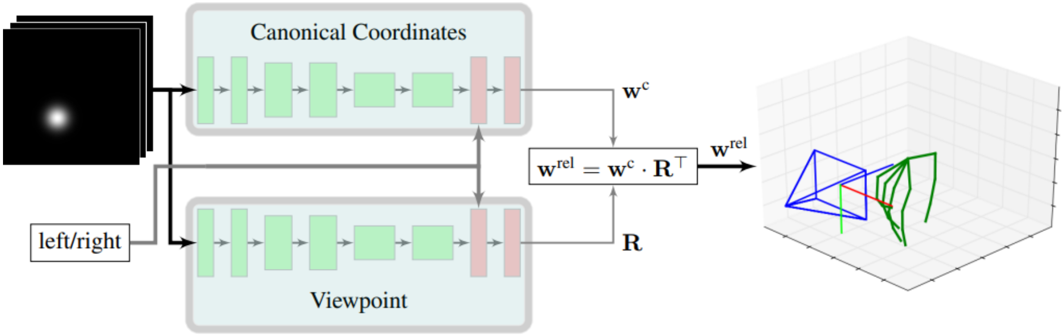



Image adopted from: Christian Zimmermann and Thomas Brox. “Learning to estimate 3d hand pose from single rgb images.” In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 4903–4911

# Generating Synthetic Annotations Using GANs

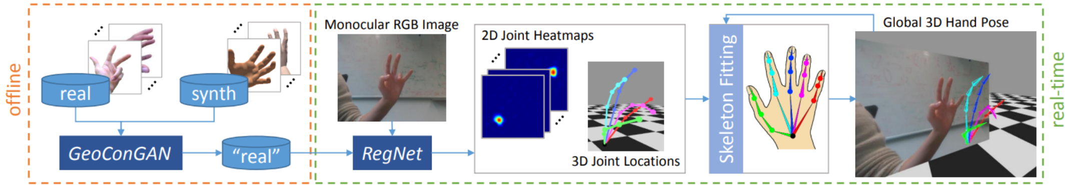



Image adopted from: Franziska Mueller et al. "GANerated Hands for Real-Time 3D Hand Tracking from Monocular RGB." In: *Proceedings of Computer Vision and Pattern Recognition (CVPR)*. 2018. URL: `https://handtracker.mpi-inf.mpg.de/projects/GANeratedHands/`

# Weakly Supervised 3D Hand Pose Estimation

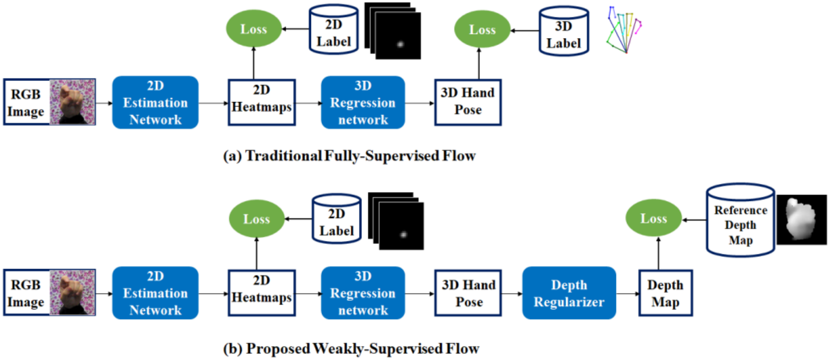



Image adopted from: Yujun Cai et al. "Weakly-supervised 3d hand pose estimation from monocular rgb images." In: *Proceedings of the European conference on computer vision (ECCV)*. 2018, pp. 666–682

# Detecting Hands Grasping Objects

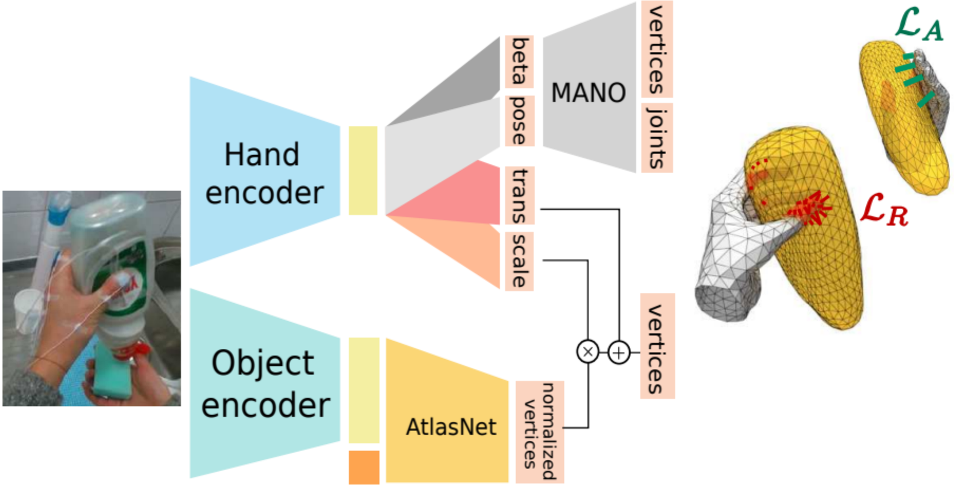
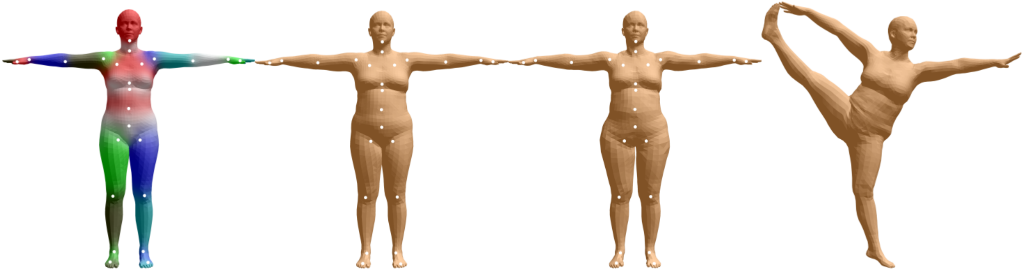



Image adopted from: Yana Hasson et al. “Learning joint reconstruction of hands and manipulated objects.” In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2019, pp. 11807–11816

# Parametric Models



(a) $\bar{\mathbf{T}}, \mathcal{W}$ (b) $\bar{\mathbf{T}} + B_S(\vec{\beta}), J(\vec{\beta})$ (c) $T_P(\vec{\beta}, \vec{\theta}) = \bar{\mathbf{T}} + B_S(\vec{\beta}) + B_P(\vec{\theta})$ (d) $W(T_P(\vec{\beta}, \vec{\theta}), J(\vec{\beta}), \vec{\theta}, \mathcal{W})$

Image adopted from: Matthew Loper et al. "SMPL: A skinned multi-person linear model." In: *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*. 2023, pp. 851–866

# SAM 3D Body

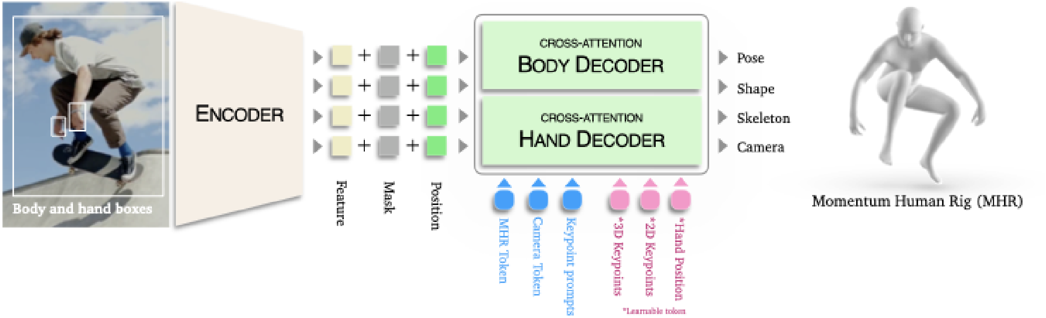



Image adopted from: Xitong Yang et al. “Sam 3d body: Robust full-body human mesh recovery.” In: *arXiv preprint arXiv:2602.15989* (2026)